Tuesday
1 Sept, 2026

Apriori Algo

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Transaction ID	Items
T1	i1, i2, i3
T2	i2, i3, i4
T3	i4, i5
T4	i1, i2, i3
T5	i1, i2, i3, i4

N = 5
min support = 3 (60 %)
min confidence = 60%.

Freq. 1-itemsets:

		Support	
i1	→ T1, T4, T5	3	
i2	→ T1, T2, T4, T5	4	
i3	→ T1, T2, T4, T5	4	
i4	→ T2, T3, T5	3	
i5	→ T3	1	X

$L_1 = \{i1\}, \{i2\}, \{i3\}, \{i4\}$

Freq. 2-itemsets:

		Support	
{i1, i2}	→ T1, T4, T5	3	
{i1, i3}	→ T1, T4, T5	3	
{i1, i4}	→ T5	1	X
{i2, i3}	→ T1, T2, T4, T5	4	
{i2, i4}	→ T2, T5	2	X
{i3, i4}	→ T2, T5	2	X

$L_2 = \{i1, i2\}, \{i1, i3\}, \{i2, i3\}$

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Q.) Find all closed & Max freq. itemsets. MinSupport = 3

Items	Appeared in	Support	
A	T1, T2, T3, T5, T6, T8	6	✓
B	T1, T3, T4, T5, T7, T8	6	✓
C	T1, T2, T4, T5, T7	5	✓
D	T2, T3, T5, T6, T7, T8	6	✓

TID	Items
T1	{A, B, C}
T2	{A, C, D}
T3	{A, B, D}
T4	{B, C}
T5	{A, B, C, D}
T6	{A, D}
T7	{B, C, D}
T8	{A, B, D}

Freq. 2-itemsets:

{A, B}	T1, T3, T5, T8	4	✓
{A, C}	T2, T5, T1	3	✓
{A, D}	T2, T3, T5, T6, T8	5	✓
{B, C}	T1, T4, T5, T7	4	✓
{B, D}	T3, T5, T7, T8	4	✓
{C, D}	T2, T5, T3	3	✓

Freq. 3-itemsets:

{A, B, C}	T1, T5	2	✗
{A, C, D}	T2, T5	2	✗
{B, C, D}	T5, T7	2	✗
{A, B, D}	T3, T5, T8	3	✓

Max = 3

= {A, B, D}

Closed:

{A, B, C, D, AB, AC, AD, BC, BD, CD, ABD}

Q.) MinSupport = 50% = 3

Beer	4
Diapers	4
Nuts	4
Coffee	2
Eggs	3
Milk	2

{Beer, Diapers} 3

T_id	Items_bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diapers, Nuts
30	Beer, Diapers, Eggs
40	Beer, Nuts, Eggs, Milk
50	Nuts, Coffee, Diapers, Eggs, Milk

Q.) Which of the following is not frequent itemset?

A. {Beer, Diapers}

~~B.~~ {Coffee}

C. {Beer}

D. {Eggs}

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